Establishing CFD Model Credibility for Hemolysis and Shear Exposure in a Centrifugal Blood Pump: ASME V&V 40–Aligned Assessment Across Three Models and Three Mechanisms

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Abstract

Computational fluid dynamics is routinely used to anticipate hemolysis risk in rotary blood pumps, yet model credibility for different damage pathways remains unevenly established. We aligned three distinct CFD approaches with the ASME V&V 40 risk-based framework and assessed their credibility against three mechanistic failure considerations: bulk hemolysis from sustained shear exposure, tip-leakage jet shear spikes, and recirculation-driven residence time in the volute and backclearance. The models comprised: (A) steady RANS using a multiple reference frame (MRF) with k–omega SST and a Newtonian analog for whole blood paired to an Eulerian shear-rate field and a power-law hemolysis post-processor; (B) transient large-eddy simulation (LES) with a sliding mesh interface and non-Newtonian Carreau–Yasuda rheology that tracked Lagrangian red-blood-cell surrogates to accumulate peak shear and exposure time; and (C) a hybrid RANS–scale-resolving detached-eddy simulation (DES) targeted to separation and leakage regions, coupled to a transported scalar for hemolysis damage and near-wall shear metrics in narrow clearances. Experiments followed ASTM F1841 in blood loops with bovine blood under clinical temperatures, supplemented by refractive-index–matched flow diagnostics including planar PIV in the volute and micro-PIV in the tip gap. We framed credibility factors on a private 0–7 scale, scored independently for every model–mechanism pair. Quantities of interest were defined so that outputs could be read against both bulk hemolysis indices and localized injury thresholds without proxy transformations. Across nine paired operating points (3–6 L/min; 2500–4000 rpm; 200–400 mmHg head), hemolysis predictions correlated with measured plasma-free hemoglobin increase rates with coefficients of 0.78 (Model A), 0.84 (Model B), and 0.82 (Model C). Mean absolute errors in normalized hemolysis indices were 28% for Model A, 21% for Model B, and 23% for Model C, each within a predeclared 30% acceptance band for bulk damage assessment. LES resolved tip-jet shear statistics closest to micro-PIV estimates (RMS velocity error 6.2% in the leakage jet), while DES with damage transport best captured residence-time–weighted shear dose maps. The credibility matrix rewarded these strengths: the highest ratings concentrated on Mechanism 2 for Model B and on Mechanism 3 for Model C. We report factor-specific scores, one-line bases, and a traceable evidence trail spanning model inputs, operating-state alignment between simulation and test, suitability of outputs, relation of the laboratory program to intended use, and software pedigree. We discuss limitations related to blood batch variability and operational handling, and we close with the implications for design decisions in the specified operating envelope.

Keywords: computational fluid dynamics, hemolysis, ASME V&V 40

1. 1. Introduction

Rotodynamic blood pumps continue to expand their role in extracorporeal life support, cardiopulmonary bypass, and temporary ventricular assistance, making red blood cell (RBC) integrity a central performance criterion in addition to hydraulic efficiency. Red cell damage results from complex interactions among shear magnitude, exposure duration, turbulence characteristics, and surface interactions in tight tolerances. Computational fluid dynamics (CFD) has become the central tool for exploring these interactions early and iteratively during design. However, without a structured framework that considers decision risk, evidence breadth, and mechanism-specific model adequacy, it is difficult to translate raw CFD outputs into supportable design or regulatory decisions. ASME V&V 40 provides such a framework by tying the required strength of evidence to the consequences of an incorrect decision and the model's influence on that decision.

We present a credibility assessment pursuant to ASME V&V 40 for a centrifugal blood pump intended for extracorporeal circulation. We evaluate three complementary CFD models against three hemolysis-related failure mechanisms that together span bulk and localized damage modes. The first model (A) is a steady-state RANS simulation in an MRF framework with k–omega SST turbulence modeling, using a Newtonian analog to represent whole blood viscosity at relevant shear rates; this approach is computationally light and common in design loops. The second model (B) is a transient large-eddy simulation with a sliding mesh approach, non-Newtonian Carreau–Yasuda rheology, and particle-based RBC surrogates used to compute extreme shear and cumulative exposure along trajectories. The third model (C) is a hybrid RANS–scale-resolving simulation using detached-eddy methods in separation and leakage regions, coupled with a transported damage scalar that approximates hemolysis accumulation in an Eulerian framework, with enhanced attention to near-wall shear in gaps and seals.

We align the modeling work with three mechanisms of interest: bulk hemolysis driven by volume-averaged sustained shear; tip-leakage jet–induced transient high shear events that disproportionately damage RBCs despite small volumetric contribution; and recirculation in the volute and backclearance that elevates residence time under moderate shear, thereby compounding sublethal damage during prolonged support. To confront these mechanisms with empirical evidence, we conducted ASTM F1841 hemolysis testing with bovine blood and high-fidelity flow mapping via particle image velocimetry (PIV) and micro-PIV in key regions, including the impeller tip gap and tongue.

Our goals are threefold: to articulate the question and context of use in decision-relevant terms; to document the computational and experimental program so that evidence can be weighed per mechanism; and to render quantitative CredScale scores for the subset of credibility factors germane to this application, resolving the scores independently for each combination of computational model and damage mechanism. In doing so, we show how the strengths of each modeling approach can be directed to the questions it answers best, and where its limitations mandate caution or supplementary evidence.

The result is a consolidated yet mechanism-aware portfolio of evidence: where the output of the model is used directly for design acceptance criteria, we require corresponding depth in the tie to measurements; where a model offers directional guidance, we record that role and restrain our claims. The sections that follow define the decision context, describe the centrifugal pump and its operating points, explain the three models and three mechanisms, and then step through risk analysis and the factors used to gauge the strength of evidence. We close with a full scoring matrix and a discussion on how this portfolio supports, qualifies, or limits sign-off decisions.

Validation match to intended use (ladder of rigor) — gradation.

a.
0: Model confronted with data from an apparatus or fluid dissimilar to the target use; operating states do not overlap.

b.
2: Data from a related configuration or surrogate fluid; partial overlap in operating space.

c.
4: Same device family and similar circuit; some operating points at target temperature and pressure.

d.
5: Same hardware and circuit; majority of the operating envelope exercised; environmental conditions matched.

e.
6: Same hardware and circuit; broad coverage of the envelope with repeat replicates and calibrated sensors.

2. 2. Question of Interest and Context of Use

The central decision question is whether the computational models provide sufficient certainty, for the intended design and evaluation tasks, to assert that hemolysis remains within pre-specified acceptance bounds across the clinically relevant operating envelope of the centrifugal blood pump. The pump is intended for extracorporeal circulation in adult patients for durations ranging from hours to a week. The use environment includes inlet venous pressures between 10 and +10 mmHg, outlet arterial pressures that vary to achieve 200–400 mmHg head, blood temperatures tightly maintained at 36–38 °C, and anticoagulation within clinical ranges. The design and evaluation tasks that depend on modeling fall into three categories: guiding blade and volute geometry to avoid harmful leakage jets; screening operating points to stay within acceptable bulk hemolysis limits; and assessing flow organization in the backclearance and volute to avoid high residence times at moderate shear.

The model influence on these tasks differs: for geometric refinement around the tip and shroud, local flow predictions exert strong influence; for operating envelope decisions about bulk hemolysis at nominal workloads, model outputs are used in conjunction with empirical trends; and for residence-time concerns, model predictions are a primary source of insight complemented by spot checks. Consequences of an incorrect conclusion are moderate to high because excessive hemolysis can cause clinical sequelae such as anemia, renal dysfunction, and consumptive coagulopathy. Accordingly, the evidence demand is tailored upward in the regions where the model is most relied upon.

We structured the operating envelope as a lattice of flow rate and speed pairs comprising 3.0, 4.5, and 6.0 L/min at 2500, 3000, 3500, and 4000 rpm, with backpressure chosen to yield heads between 200 and 400 mmHg. The primary acceptance criterion for bulk hemolysis is that the normalized index of hemolysis (NIH) under ASTM F1841 remains below 0.02 g/100 L when corrected for hematocrit and plasma-free hemoglobin baseline, with a safety margin to account for patient variability. For localized damage induced by tip-leakage jets, we framed an engineering threshold of 150 Pa for instantaneous RBC-scale shear stress, and for sublethal accumulation during recirculation, a residence-time–weighted shear dose of 50 Pa·s was used as a screening criterion developed from literature and internal data. The modeling task is to produce outputs that can be checked directly against these criteria or traced to them without opaque conversions.

The benchtop program used the same pump hardware, circuit elements, and working temperature as intended in practice, and it exercised most of the operating states used in decision-making, such that 95% of the in-scope combinations of flow, speed, and head were represented in the laboratory program.

Figure 1: Schematic of the centrifugal blood pump and principal regions of interest for hemolysis assessment: inlet eye, impeller blades, tip clearance, volute tongue, backclearance, and outlet.

Model influence and decision consequence (ladder of risk tiering) — gradation.

a.
0: Model is advisory; incorrect conclusions carry negligible safety or performance impact.

b.
2: Model provides design insight; decisions cross-checked with robust testing; incorrect conclusions have low impact.

c.
4: Model guides design choices and operating envelopes; incorrect conclusions have moderate impact mitigated by testing.

d.
5: Model is a primary input to accept/reject decisions; incorrect conclusions have moderate to high impact; mitigations exist.

e.
6: Model is critical to acceptance; incorrect conclusions carry high impact; mitigations limited.

3. 3. Device Description and Operating Conditions

The device is a magnetically coupled centrifugal blood pump with a shrouded impeller of 50 mm outer diameter, seven backward-swept blades with 3.5 mm span, and a design shaft equivalent speed of 3000 rpm at 4.5 L/min and 300 mmHg head. The tip clearance between blade tip and stationary shroud is nominally 150 micrometers, with manufacturing tolerance of ±20 micrometers. A backclearance between the impeller back shroud and the pump housing forms a leakage path with 300 micrometers nominal gap. The volute cross-section has a progressive area increase to maintain near-constant circumferential velocity, with a tongue of 1.2 mm minimum tip gap to the impeller periphery. Flow enters axially through a beveled inlet eye with a hydrodynamically contoured stator hub; secondary flows in the eye can affect recirculation during low-flow or high-head operation.

The pump was tested and modeled across a matrix of setpoints: 3.0, 4.5, and 6.0 L/min volumetric flow rates at 2500, 3000, 3500, and 4000 rpm, yielding 12 combinations of flow and speed. At each combination, we adjusted downstream resistance to achieve heads in the 200–400 mmHg range. Blood temperature was maintained at 37.0 ± 0.3 °C. Hematocrit for hemolysis testing varied between 38% and 42% across runs, with target 40%. Anticoagulation used heparin at clinically relevant concentrations, monitored to avoid clotting in the loop. Sensors included flow meters calibrated to within 1.0% uncertainty, pressure transducers within 1.0 mmHg, and thermistors within 0.1 °C. The test loop conformed to ASTM F1841 geometry except where dictated by the pump's specific connections and required priming volumes.

Hydraulic performance maps confirmed the expected head–flow relationships and indicated a design point at 4.5 L/min and 300 mmHg head at 3000 rpm that we reference later as a renamed operational setpoint, OP-4.5-300, used to anchor time-resolved and steady simulation comparisons. In steady conditions, typical Reynolds numbers based on impeller tip speed and gap height exceeded 20,000, while local shear rates in the tip clearance approached 100,000 s1. These values motivate attention to both mean and fluctuating shear components for a realistic assessment of RBC damage, since time-resolved peaks can exceed hemolytic thresholds even when time-averaged stresses remain moderate.

4. 4. Computational Models: A (RANS–MRF), B (LES–Sliding Mesh), C (Hybrid DES + Damage Transport)

Model A: Steady RANS–MRF (SST). The first model uses a segregated steady-state solver with a multiple reference frame (MRF) treatment for the rotating impeller region. Turbulence closure is provided by the k–omega shear-stress transport (SST) model to balance near-wall and free-stream behavior. The fluid is represented as a Newtonian analog with dynamic viscosity set to 3.5 mPa·s, consistent with whole blood behavior at shear rates above approximately 500 s1, which dominate in the pump's energetic regions. The computational mesh comprises 12.6 million polyhedral cells with prism layers to achieve y+ values between 1 and 2 on most wetted surfaces, and cell sizes in the tip gap of 10 micrometers to resolve the shear layer thickness under MRF approximations. The impeller, shroud, and volute are meshed in a single connectivity with frozen rotor interfaces at the region boundaries. A converged solution is obtained when residuals drop below 1e-5 and monitor points for torque and integral flow rates stabilize within 0.2% over 500 iterations. Shear-rate fields are recorded and passed to a hemolysis index post-processor implemented as H = C × τ^α × t^β, with constants C = 3.62e7, α = 2.416, and β = 0.785, applied to elementwise τ and an exposure time estimated from local convection scales.

Model B: Transient LES–Sliding Mesh. The second model solves the unsteady, filtered Navier–Stokes equations with a dynamic Smagorinsky subgrid-scale model, coupled to a non-Newtonian Carreau–Yasuda rheology for blood: μ(γ) = μ + (μ0 μ)[1 + ($\lambda\gamma$)^a]^(n1)/a, where μ = 3.5 mPa·s, μ0 = 0.16 Pa·s, λ = 3.313 s, a = 2.0, and n = 0.3568. The mesh contains 81 million cells with near-wall spacing that yields target y+ 1 for LES wall modeling and 6 micrometer minimum spacing in the tip clearance. A conformal sliding interface separates the rotating impeller from stationary domains to capture blade-passing effects. Time steps are chosen to satisfy a Courant number below 0.5, typically Δt = 1.0e5 s, with simulations run for at least 20 impeller revolutions to reach statistical stationarity before accumulating 40 revolutions of data for statistics. RBC surrogates are represented by 100,000 Lagrangian particles per run, neutrally buoyant with an effective diameter of 8 micrometers; Stokes drag dominates, but additional forces from pressure gradients and added mass were included to confirm negligible impact on path statistics. Each particle records instantaneous shear stress along its path, enabling computation of peak shear and exposure time exceeding thresholds, and, when desired, accumulation of the same power-law hemolysis index for comparison to bulk measures.

Model C: Hybrid RANS–Scale-Resolving (DES) with Scalar Damage Transport. The third model applies delayed detached-eddy simulation (DDES) in regions of expected separation, leakage, and strong curvature while retaining RANS in attached boundary layers. The mesh has 35 million cells with local refinement to 8 micrometers in the tip gap and near the volute tongue. Within the DES regions, a k–omega base closure morphs to LES-like behavior depending on grid and flow conditions. A transported scalar, D(x,t), approximates cumulative RBC damage, initialized to zero at the inlet and convected with the flow, with production S_D proportional to τ^α, where τ is the second invariant of the strain-rate tensor scaled by viscosity, using the same exponents α and β used in post-processing for Models A and B but embedded in a transport equation with effective diffusion set to a small fraction (1%) of the molecular value to model dispersion. Near-wall shear metrics are also recorded in the narrow clearance regions. Time advancement uses dual time-stepping to converge pseudo-transients per physical time step of 5.0e5 s over 30 impeller revolutions for statistical averaging.

All models were implemented in the same commercial CFD platform with consistent numerics: second-order spatial discretization for convection terms using a bounded central differencing scheme for LES-like regions and a high-resolution upwind blend for RANS regions; pressure–velocity coupling through a coupled solver for unsteady runs and SIMPLEC for steady runs; and pressure–staggering to maintain robustness at high rotation rates. For each model, grid and time-step sensitivity studies were conducted at OP-4.5-300 and at a low-flow high-head case to probe residence-time behavior. The selected grids and time steps achieved less than 5% change in integral hemolysis indices upon further refinement.

The computational costs scaled with fidelity: Model A converged in about 6 hours on 160 CPU cores; Model B required approximately 48 hours on 512 cores for 60 revolutions; Model C fell in between, at roughly 24 hours on 256 cores for 30 revolutions. These costs informed where each model was deployed within the design and evaluation workflow, with the steady MRF model used for screening, LES for detailed diagnostics especially in tip jets, and DES with scalar transport for residence-time mapping.

Pedigree of boundary and material data (ladder of rigor) — gradation.

a.
0: Boundary values and fluid properties taken from textbooks or broad literature without traceability.

b.
2: Boundary values measured once for the campaign; properties taken from literature with nominal corrections.

c.
4: Boundary values and temperature measured per run; viscosity fitted to a standard curve; limited uncertainty quantification.

d.
5: All boundary values measured with calibrations; non-Newtonian coefficients estimated from in-run samples; uncertainties documented.

e.
6: Boundary and property data measured run-by-run with calibrated instruments; 95% intervals propagated in ensembles.

5. 5. Hemolysis and Shear Exposure Mechanisms Assessed

Mechanism 1: Bulk hemolysis from sustained shear exposure. In rotary blood pumps, the majority of RBC damage arises from time spent in moderate-to-high shear zones throughout the impeller and volute. While instantaneous shear spikes can be severe locally, the volumetric contribution of persistent shear is the primary driver of plasma-free hemoglobin accumulation in most operating conditions. This mechanism is suitably represented by a power-law damage model that integrates local shear and exposure time across the volume, producing a hemolysis index that can be confronted with normalized hemolysis indices (NIH) obtained from loop testing. Model A directly targets this mechanism through an Eulerian field with a scalar damage post-processor that assumes exposure time based on convection scales; Models B and C can also generate bulk hemolysis predictions by integrating along paths (B) or integrating the transported scalar (C).

Mechanism 2: Tip-leakage jet shear spikes. As fluid leaks through the small gap between the impeller blade tip and the shroud, a high-velocity jet forms that can reach several tens of meters per second locally, with shear layers that attain instantaneous stresses well above 150 Pa depending on gap, speed, and pressure differential. Although the total flow in the tip gap is a small fraction of the pump throughput, RBCs entering this region can experience acutely damaging stress-time histories. Capturing this mechanism requires resolving the unsteady jet and its shear layers; Model B's time-resolved LES with a sliding mesh is well-suited to this, while Model C's DES partially resolves the dynamics in leakage paths. Model A's steady MRF can reconstruct mean fields but is less effective at representing unsteady peak stresses that set the damage for cells that happen to pass at unfavorable phases.

Mechanism 3: Recirculation-driven residence time in volute and backclearance. Accumulation of sublethal damage occurs when RBCs spend extended periods in moderate shear, particularly in recirculation pockets near the volute tongue and in the backclearance region behind the impeller. This accumulation can sensitize cells to later insults and can, over long support durations, contribute to aggregate hemolysis beyond what instantaneous thresholds would suggest. Quantifying residence time requires either particle tracking to estimate distributions of path durations (Model B) or an Eulerian proxy based on scalar transport with small diffusivity (Model C). Model A can estimate the presence and extent of recirculation zones and magnitudes of mean shear but cannot represent the time distribution of exposure without additional modeling assumptions.

The outputs were defined to align directly with what is measured and what underlies safety thresholds. Specifically, the hemolysis index used in computations adopts the same power-law form and parameters as used to interpret ASTM F1841 loop results (C = 3.62e7, $\alpha = 2.416$, $\beta = 0.785$) with an a priori acceptance of ±30% mean absolute error; the maximum instantaneous shear accumulated along tracked paths is compared to a 150 Pa injury threshold; and a residence-time–weighted shear dose threshold of 50 Pa·s is used to screen regions and operating points for sublethal accumulation risk.

Traceability of outputs to decision metrics (ladder of rigor) — gradation.

a.
0: Outputs are proxies with no clear relationship to decision thresholds.

b.
2: Outputs correlate qualitatively with decision metrics; scaling factors required.

c.
4: Outputs map via empirical conversion to measured indices; documented calibration exists.

d.
5: Outputs defined with the same units and definitions as measurements; minor post-processing harmonization.

e.
6: Outputs are identically defined to measurements; no conversion required; thresholds predeclared.

6. 6. Model Risk Analysis (ASME V&V 40): Influence, Decision Consequence, Overall Risk and Rationale

Mechanism-specific risk classification anchors the required strength of evidence. For Mechanism 1 (bulk hemolysis), model outputs steer acceptance of the operating envelope in conjunction with empirical data. Because incorrect acceptance could lead to elevated hemolysis in patients, the decision consequence is high, but reliance on the model is moderated by parallel loop testing at representative points. We classify model influence as medium and decision consequence as high, suggesting a high but not absolute demand for evidence depth, with cross-validation required. For Mechanism 2 (tip leakage spikes), model outputs directly inform micro-geometry and tip gap targets, and empirical confirmation is constrained by the difficulty of measuring shear in situ, despite micro-PIV providing useful kinematics. Here, the model's influence is high while decision consequence is moderate to high. Evidence depth for the jet dynamics and peak shear statistics is therefore set high, particularly for the time-resolved models. For Mechanism 3 (recirculation residence time), decisions about shroud contouring and backclearance rely heavily on modeling because benches measure aggregate hemolysis but not distributed residence times, though dye-washout experiments can partially corroborate. Influence is high, consequence moderate to high. The resulting evidence demand favors models that directly compute residence-time distributions or scalar analogs with quantifiable dispersion.

Within this framework, Model A is expected to be adequate for Mechanism 1 when bulk behavior dominates, but less suitable to set micro-geometric decisions about tip jets, where steady approximations underrepresent maxima. Model B should achieve the highest trust for tip leakage and solid performance for bulk hemolysis provided sufficient sampling and averaging are performed. Model C, by virtue of a transported damage scalar, is a strong candidate for diagnosing residence-time–weighted dose hotspots; its ability to represent instantaneous tip shear will be somewhat limited relative to LES. These expectations informed the weight we placed on each model's outputs in decision deliberations and the performance criteria we established for each model–mechanism pair in validation.

Boundary values and material properties used in the simulations were drawn from the same test-day measurements that governed the bench runs, with 95% confidence intervals carried through in 300-member ensembles per representative condition; variation in hematocrit (38.5 ± 1.2%) and viscosity at 100 s1 (3.8 ± 0.2 mPa·s) were explicitly sampled, and input uncertainty accounted for 12–18% of the spread in predicted hemolysis across the three models.

Alignment of operating points between simulation and test (ladder of rigor) — gradation.

a.
0: Operating points not matched; differences exceed 10% in speed or flow.

b.
2: Nominal setpoints matched; actual conditions differ by up to 10%.

c.
4: Actual conditions matched within 5% for speed and flow; temperature within 1 °C.

d.

5: As at level 4 with pressure head matched within 5% and viscosity within 10%.

e.
6: Actual speed and flow matched within 2%; head within 3%; viscosity within 5%; gap measured or bounded.

7. 7. Software Quality Assurance: Configuration control, solver audits, and regression checks

All simulations were executed on a controlled software baseline installed on two high-performance computing clusters and a workstation pool for pre- and post-processing. Configuration management used a distributed version control system with protected branches and peer review for all merge requests. Solver binaries and libraries were hashed and archived per build, and run scripts were templated with version tags and environmental fingerprints. The project maintained a catalog of third-party dependencies, including compiler versions and math libraries, to ensure reproducibility.

A structured test suite was used to verify numerical integrity after each quarterly update to the CFD platform or when migrating between clusters. The suite included canonical flows relevant to rotary devices: laminar Couette flow for shear stress verification, turbulent channel flow with known friction coefficients, Taylor–Couette instability onset thresholds, and a centrifugal pump surrogate at low head where quasi-analytic performance exists. For each case, tolerances on power number, skin friction, and velocity profiles were set from published benchmarks and internal references. Test execution was automated and logged with pass/fail status, summary statistics, and archived residual histories. Beyond the automated suite, periodic solver audits were conducted by a separate computational group to replicate key study results from raw meshes, boundary definitions, and scripts.

Numerical settings were documented and enforced through templates to avoid silent changes: discretization schemes, convergence criteria, relaxation factors for steady runs, and time-step controllers for unsteady runs were all parameterized and recorded. Mesh generation scripts were versioned, with meshing parameter files tied to geometry revisions via a shared key. Geometry models were maintained under product lifecycle management with change orders and approvals linking to both experimental and computational work packages. The data management approach extended to experimental datasets for alignment: dashboards mapped CFD conditions to test runs with metadata, ensuring traceability.
Software and workflow pedigree (ladder of rigor) — gradation.

a.
0: Ad hoc workflows; no version control; runs not repeatable.

b.
2: Informal version control; some key settings documented; partial repeatability.

c.
4: Formal version control; automated unit tests; documented numerics; change logs maintained.

d.
5: Regression test suite relevant to application; archived binaries; cross-platform checks.

e.
6: Comprehensive regression suite; independent audits; full provenance of binaries and inputs; automated pass/fail gates.

8. 8. Model Inputs: Blood rheology, boundary conditions, rotational speed, and renamed operational setpoint

The simulations used boundary conditions derived from the operating envelope. Inlet conditions were defined with a fixed total pressure set to match venous conditions in the loop, typically near 0 mmHg gauge with excursions depending on target head. Outlet conditions were prescribed as a static pressure adjusted to match the desired head for each flow–speed pair. Rotational speed was applied as a frame speed (MRF) in Model A and as an angular velocity on the rotating region in Models B and C. The renamed benchmark setpoint, OP-4.5-300, refers to 4.5 L/min at 3000 rpm with approximately 300 mmHg head, used as a recurring anchor for sensitivity and validation tasks.

Blood was represented by different rheological models depending on the computational approach. Model A used a Newtonian analog with dynamic viscosity of 3.5 mPa·s—a value consistent with high-shear behavior of blood at hematocrit ~40% and 37 °C, reflecting the dominance of elevated shear in the impeller and tip gaps. Models B and C used the Carreau–Yasuda constitutive law with parameters chosen to reproduce the shear-thinning behavior measured in the laboratory viscometer. Density was set to 1050 kg/m^3 across all models. Temperature was set to 37 °C and considered isothermal, consistent with loop control and the minor Joule heating in the test durations.

Flow development and turbulence intensities at the inlet were based on measurements upstream of the pump's inlet cone in the loop, represented as low-level turbulence intensity (~5%) and length scale linked to the inlet diameter. Surface roughness was assumed negligible relative to viscous sublayer thickness, consistent with polished polymer and metal surfaces used in the pump and loop.

For Models B and C, time histories were collected over many impeller revolutions to develop adequate statistics on unsteady quantities. Particle injections in Model B occurred over multiple blade phases to represent uniform phase sampling, and injection positions were distributed across the inlet cross-section with a slight weighting toward the hub and shroud to account for measured secondary flows in the inlet eye. Model C's damage scalar was initialized to zero and monitored at the outlet and in planes cutting through the volute, tongue, and backclearance to understand the spatial accumulation of dose.

9. 9. Equivalency of Input Parameters Across Models and Test Conditions

To limit confounding differences between simulation and laboratory conditions, we sought close alignment of operational states and material properties. This included mapping each computational run to a specific bench run, using the measured rotational speed, measured flow rate, measured outlet pressure (to define head), and day-specific temperature. In models that required additional inputs such as non-Newtonian parameters, we used coefficients inferred from the same blood batch or from bracketed batches in time. For micro-geometric features, we measured a sample of pumps for tip gap and backclearance using bore gauges and endoscopes, and we used the sample mean as the baseline geometry in the computational meshes while exploring sensitivity to ±1 standard deviation. Sensors in the loop were calibrated before runs, and correction factors were applied to reconstruct best estimates of true operating points. The computational meshes reflected as-manufactured nominal geometry except for the gap parameterization.

Across the three modeling approaches, a common set of geometric models was used to the extent allowed by solver-specific meshing constraints, and care was taken to align boundary plane locations so that flow rate and pressure comparisons were not biased by section choices. For Model B's sliding interface, the location of the interface was chosen to coincide with a plane where the MRF interface was located in Model A, and for Model C, DES regions were defined with the same extents as the LES high-resolution zones in Model B's grid. All CFD cases included the hydraulic resistance of the outlet pipe segment modeled to the same distance as in the bench loop to harmonize head–flow relationships. Where inlet turbulence intensity was required, we used measurements to set comparable levels across models.
Cross-environment parameter matching (ladder of rigor) — gradation.

a.
0: Simulation and experiment use dissimilar parameter sets; key values are unmatched.

b.
2: Major parameters (speed, flow) matched nominally; minor properties differ.

c.
4: Major parameters matched to within specified tolerances; properties harmonized via literature.

d.
5: As at level 4 with day-specific temperature and pressure matches; partial geometry checks.

e.
6: All principal parameters matched to tight tolerances; geometry tolerances bounded; consistent boundary planes.

10. 10. Quantities of Interest and Acceptance Criteria: Hemolysis index, peak shear exposure, residence-time–weighted shear

To support mechanism-specific judgments, we defined outputs so that they could be aligned with measurements and thresholds. For Mechanism 1, bulk hemolysis is characterized by a hemolysis index proportional to the integral of a shear stress power law over time. For Model A, we computed the hemolysis index by applying the Giersiepen law at each cell using the Eulerian shear-rate magnitude and an exposure time linked to path length and local velocity; the pump-wide index is the volume integral normalized by throughput. For Model B, an equivalent hemolysis index is accumulated along each Lagrangian particle path, with per-particle values binned to yield a population distribution; the bulk index is the mean or a selected percentile of this distribution. For Model C, the transported damage scalar gives a spatial map whose outlet-averaged value supplies a bulk index analogous to the loop measurement.

For Mechanism 2, the quantity of interest is the distribution of maximum instantaneous shear along RBC surrogate trajectories. We collect the empirical cumulative distribution function (ECDF) of peak shear per particle and report its percentiles. This distribution is confronted with an injury threshold of 150 Pa; the fraction of particles exceeding this threshold is a diagnostic of acute damage risk in the tip gap and volute tongue regions. For Model A, which lacks time resolution, we used a proxy based on twice the local shear stress as an estimate for high-frequency fluctuations and noted its limitations when interpreting comparisons to measurements.

For Mechanism 3, the residence-time–weighted shear dose is taken as the integral of shear stress over the time a particle spends above 25 Pa but below 150 Pa; this selects the moderate range most implicated in sublethal accumulation. Model B supplies this directly from particle path histories. Model C's scalar offers an Eulerian analog by integrating production in regions where shear lies in that band, with a small dispersion term to reflect mixing. We summarize the risk via statistics of this dose across the outlet flow or spatial maps highlighting hotspots in the backclearance and volute tongue regions.

Acceptance criteria were established a priori. For bulk hemolysis, model predictions at the nine designated operating points are expected to achieve mean absolute error under 30% relative to measured normalized hemolysis indices and to correctly identify whether a case lies above or below the NIH 0.02 g/100 L threshold. For tip-jet–induced peaks, the model is expected to identify less than 1% of pathlines exceeding 150 Pa at OP-4.5-300 and to diagnose any settings where that fraction exceeds 3%, triggering design review. For residence-time dose, the model should show that 95% of the outlet flow has cumulative dose under 50 Pa·s at OP-4.5-300; if not, flow reorganization features are warranted.
Suitability of outputs and targets (ladder of rigor) — gradation.

a.
0: Outputs are loosely connected to targets; thresholds unspecified.

b.
2: Outputs share units with targets; thresholds defined post hoc.

c.
4: Outputs and thresholds defined a priori; mapping requires model-specific transformations.

d.
5: Outputs and thresholds defined a priori with harmonized definitions; limited conversion.

e.
6: Outputs identical to measurement definitions; thresholds predeclared; reporting includes distributional checks.

e.
6: Comprehensive diagnostics targeting mechanism-defining features; repeated over operating states.

11. 11. Relevance of Validation Activities to the COU: In vitro hemolysis per ASTM F1841 and flow diagnostics

We executed an in vitro program comprising hemolysis testing per ASTM F1841 and flow diagnostics designed to interrogate key velocity and shear features of the pump. Hemolysis tests used fresh bovine blood procured on the morning of testing, anticoagulated with heparin, and filtered per protocol. The loop incorporated the subject pump, a heater–cooler to maintain temperature, a reservoir, and downstream resistances to set pressure head. Samples were drawn at fixed intervals over 120 minutes, plasma separated, and plasma-free hemoglobin quantified spectrophotometrically. Hematocrit and plasma protein content were recorded to correct NIH per the standard. Each operating point was repeated at least three times to characterize run-to-run variability.

Flow diagnostics included planar PIV in a water–glycerol fluid matched in refractive index to transparent pump housings made for this purpose. Seeding density and light sheet thickness were tailored to the volute and impeller passages. Velocity fields were reconstructed from paired images using multi-pass cross-correlation with window deformation and validation lights. Micro-PIV focused on the tip gap: a microfabricated analog of the tip region with a controllable gap replicated the hydrodynamic environment, enabling measurements of leakage jet velocities and shear layer thicknesses. These diagnostics, while not performed in blood, were designed to capture the kinematics that set shear exposure.

Supplementary dye-washout experiments in the backclearance and volute provided qualitative residence-time information to complement the model's scalar or particle-based metrics. We introduced a fluorescent dye pulse upstream of the backclearance and recorded intensity decay with high-speed cameras at fixed locations, extracting exponential time constants as a coarse benchmark for model-predicted residence times. These activities were planned alongside the computational program so that regions of high interest in the models were deliberately probed in the experiments whenever feasible.

Validation breadth for targeted mechanisms (ladder of rigor) — gradation.

a.
0: Single-point validation unrelated to target mechanism.

b.
2: Limited data near the region of interest; different fluid or geometry.

c.
4: Multiple data sets probing some aspects of the mechanism; partial overlap with model regions.

d.
5: Rich data set on kinematics in the mechanism region; direct measurements of related quantities.

12. 12. Test Samples: Pump builds and blood analog preparation (noting practical constraints)

The pump builds used in the hemolysis tests were drawn from production-intent components assembled under clean-room conditions. Assemblies followed standard torque specifications and clearances validated against manufacturing records. Each test article underwent basic hydraulic performance verification before use in the loop to ensure function within expected head–flow bounds. Blood was procured from a single regional supplier in compliance with institutional protocols, transported under refrigeration, and processed on the day of testing. Blood age, storage conditions, and batch identifiers were logged alongside hematology measures taken before and after tests.

For the optical analogs, transparent housings were fabricated from optically clear polymers and were dimensionally checked against the metal pump housings; minor differences in surface energy and wetting behavior were acknowledged in planning. The water–glycerol working fluid used for PIV was prepared to match the refractive index of the polymer within the limits of the temperature-controlled environment to minimize optical distortion. Seeding particle concentration and size distributions were chosen to balance tracking fidelity against signal quality. While these preparations emulate relevant conditions for flow visualization, we note that they represent a compromise between optical access and exact duplication of blood rheology.

We coordinated sample scheduling to minimize temporal drift in blood properties during the test day and arranged the order of operating points to match computational runs prepared in advance. Between tests, the loop was flushed and re-primed following a standard operating procedure aimed at consistent handling. Although the emphasis was on maintaining continuity across runs, variations inherent in biological samples and in mechanical handling were present and recorded for later analysis.

Characterization of physical test articles and fluids (ladder of rigor) — gradation.

a.
0: No characterization of test articles or fluids; unknown provenance.

b.
2: Basic identification and logging; limited checks of geometry or fluid properties.

c.
4: Batch-level characterization; geometry checked against drawings; fluid properties measured before testing.

d.
5: Per-specimen measurements of key dimensions and fluid hematology; controlled storage and handling.

e.
6: Full lot tracking; per-specimen gap metrology;

rheology characterized across shear rates; handling uniformity assessed.

13. 13. Use Error: Operational handling, priming, and data acquisition considerations

Operational handling in both computational and experimental workflows can introduce discrepancies that are not attributable to model form or physics. In the loop, priming procedures, elimination of air, alignment of connectors, and positioning of pressure taps all influence the repeatability of head–flow readings and the likelihood of transient events that can seed hemolysis. In data acquisition, timing of sample draws and scrambling of operator actions across runs are common sources of variability. On the computational side, meshing choices, smoothing of sharp edges, and the placement of interface planes between rotating and stationary domains influence resolved shear and the representation of leakage.

To mitigate these aspects, we used detailed standard operating procedures, checklists, and synchronized time stamps across sensors and sample draws. Operators were briefed before each test block and debriefed afterward to record deviations. Computational operators used meshing templates and pre-run checklists, including checks for gap resolution and y+ targets. Post-processing scripts were validated against unit tests to avoid analysis drift over time. A subset of runs included independent oversight by a quality officer who was not part of the day-to-day team, focusing on adherence to procedures. The integration of this oversight with the technical program provided context for observed variability.

Despite these controls, unavoidable human-in-the-loop elements remained, especially in reconciling pressure drops across fittings and in establishing identical pre-fill conditions across loop resets. By logging these elements and their potential impacts, we preserved the ability to interpret outliers in the hemolysis data or in flow diagnostics in light of operational realities.

Operational controls against handling-induced deviations (ladder of rigor) — gradation.

a.
0: No documented procedures; operator-dependent; unlogged deviations.

b.
2: Basic SOPs; limited training; sporadic logging of deviations.

c.
4: Detailed SOPs and checklists; training conducted; deviations logged and reviewed.

d.
5: Formal human factors review; independent oversight on a sample of runs; synchronized data acquisition.

e.
6: Comprehensive human factors engineering; mock runs; pre-brief and debrief documented; drift monitoring.